National University of Computer & Emerging Sciences, Karachi

Fall-2026 School of Computing (BSCY)

Assignment # 1

Subject: Database Systems	**Post Date: 1/09/2026**
Total Marks: 50	**Due Date: 13/09/2026**
Course Instructors: Ms. Abeer Gauher	

Instructions to be strictly followed.

- All the submissions will be made on Google Classroom.
- It should be clear that your assignment would not get any credit if the assignment is submitted after the due date.
- Zero grade for plagiarism (copy/ cheating) and late submissions.

Question One: 20 marks

A university wants to develop a database system to manage its course registration process.

The university has several departments. Each department has a unique department ID, name, office location, and contact number. A department offers multiple courses and employs several instructors.

Each course has a unique course ID, course title, credit hours, and course level. Every course belongs to one department. Some courses may have prerequisites, meaning that a student must successfully complete another course before registering for that course.

The university has several instructors. Each instructor has an instructor ID, name, email address, phone number, designation, and salary. Every instructor belongs to one department.

The university also has students. Each student has a unique student ID, name, date of birth, email address, phone number, admission date, and academic status. Every student belongs to a department.

Courses are offered during different semesters. For each course offering, the university needs to store the semester, academic year, section number, classroom, schedule, and the instructor responsible for teaching the section.

A student can register for multiple course sections, and each course section can have multiple students. For every registration, the university wants to record the registration date, enrollment status, and final grade.

Tasks

Task 1: Carefully analyze the given scenario and identify all the major entities required to design the database.

For each entity, provide:

- Entity name
- Brief description of the entity

Task 2: For every identified entity, list all relevant attributes required to store its information.

Task 3: For every entity, identify an appropriate Primary Key (PK).Your answer should explain:

- Which attribute is selected as the primary key
- Why it uniquely identifies each record.

Task 4: Identify all attributes that should act as Foreign Keys (FKs).For each foreign key, specify:

- Foreign key attribute
- Referenced table
- Referenced primary key

Task 5: For every attribute, specify an appropriate domain. The domain should indicate the type and restrictions of values that the attribute can contain.

Consider:

- Data type
- Maximum length where applicable
- Range of values
- Whether NULL values are allowed

Task 6: Convert your analysis into a relational database schema clearly showing the keys.

Question Two: 30 marks

A university wants to maintain information about its departments, students, instructors, courses, course sections, and student registrations. The database contains the following tables:

Department: Stores information about university departments.

- DepartmentID
- DepartmentName
- OfficeLocation
- ContactNumber

Student:Stores information about students.

- StudentID
- StudentName
- DateOfBirth
- Email
- Phone
- AdmissionDate
- AcademicStatus
- DepartmentID

Instructor: Stores information about instructors.

- InstructorID
- InstructorName
- Email
- Phone
- Designation
- Salary
- DepartmentID

Course: Stores information about courses offered by departments.

- CourseID
- CourseTitle
- CreditHours
- CourseLevel
- DepartmentID

Course Section: Stores information about individual sections of courses offered during a semester.

- SectionID
- CourseID
- InstructorID
- Semester
- AcademicYear
- SectionNumber
- Classroom
- Schedule

Registration: Stores information about students registering for course sections.

- RegistrationID
- StudentID
- SectionID
- RegistrationDate
- EnrollmentStatus
- FinalGrade

1. Write SQL CREATE TABLE statements for all six tables. Your tables must include:
    - Appropriate data types
    - Primary keys
    - Foreign keys
    - NOT NULL constraints where appropriate
    - UNIQUE constraints where appropriate
    - CHECK constraints where appropriate
    - Ensure that the foreign keys correctly establish the relationships between the tables.

2. Write an SQL statement using ALTER TABLE to add: City VARCHAR(50) to the Student table.
3. Write an SQL UPDATE query to modify the student's academic status.Your query should identify the student using their StudentID.
4. A course section has been cancelled before any students registered for it. Write an SQL DELETE statement to remove the relevant course section.
5. Write a query to display the following information for all students: Student ID,Student Name,Email, Academic Status. Display the results alphabetically according to the student's name.
6. Write a query to find: Highest instructor salary, Lowest instructor salary, Average instructor salary
7. Write a query to determine how many students belong to each department. The output should display: Department ID, Number of student. Sort the results by the number of students in descending order.
8. Write an SQL query to display: Student ID, Student Name, Department Name for every student.
9. Write a query using multiple joins to display: Student Name, Course Title, Instructor Name, Semester, Enrollment Status, Final Grade for every registration.
10. Write a query to find students who have not registered for any course section.Display: Student ID, Student Name.
11. Write a query using a subquery to find all students who belong to the department named "Computer Science".Display: Student ID, Student Name, Email
12. Write a query using a subquery to find instructors whose salary is greater than the average salary of all instructors. Display:Instructor ID, Instructor Name, Salary
13. Write a query to display courses whose credit hours are greater than the average credit hours of all courses.
14. Write a query using a subquery to find the names of students who are registered in a course titled "Database Systems.
15. Write a query that displays each department along with: Department name, Number of students, Number of courses, Average instructor salary. Only display departments where the average instructor salary is greater than the overall average instructor salary.